Unsupervised Learning

Unsupervised Learning means **learning from data without labeled outputs**.

There is **no target variable (Y)**

Age	Income	Married	Gender
34	56000	y	male
39	70000	n	female
28	23000	n	male
42	49000	y	male

Unsupervised Learning - Types

Clustering

Dimensionality Reduction

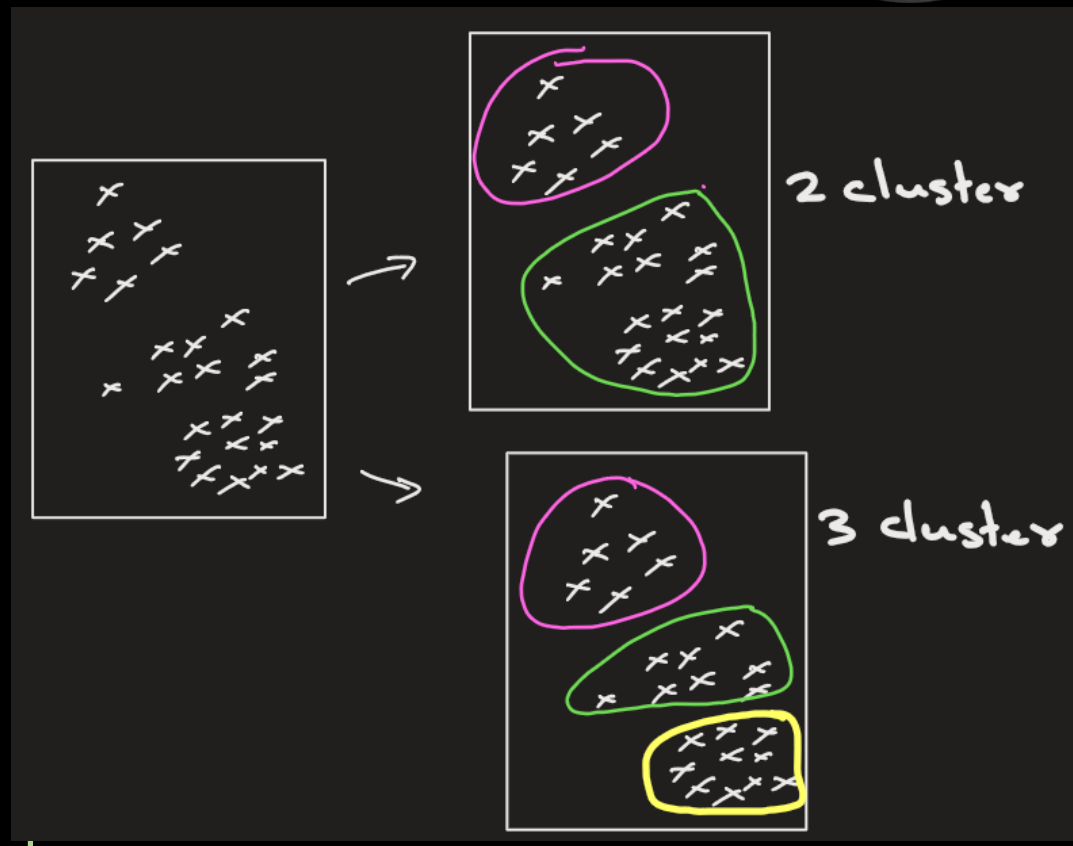
Association Rules

Anomaly Detection

Generative Models

Types

Clustering:
Groups similar data points together and find patterns



Common Algorithms:
- K-Means
- Hierarchical Clustering
- DBSCAN
- Mean Shift

Example:
- Customer segmentation: High spenders vs Low spenders
- Grouping similar documents: Group hospital bills and doctor report separately
- Image segmentation

Types

Dimensionality Reduction:

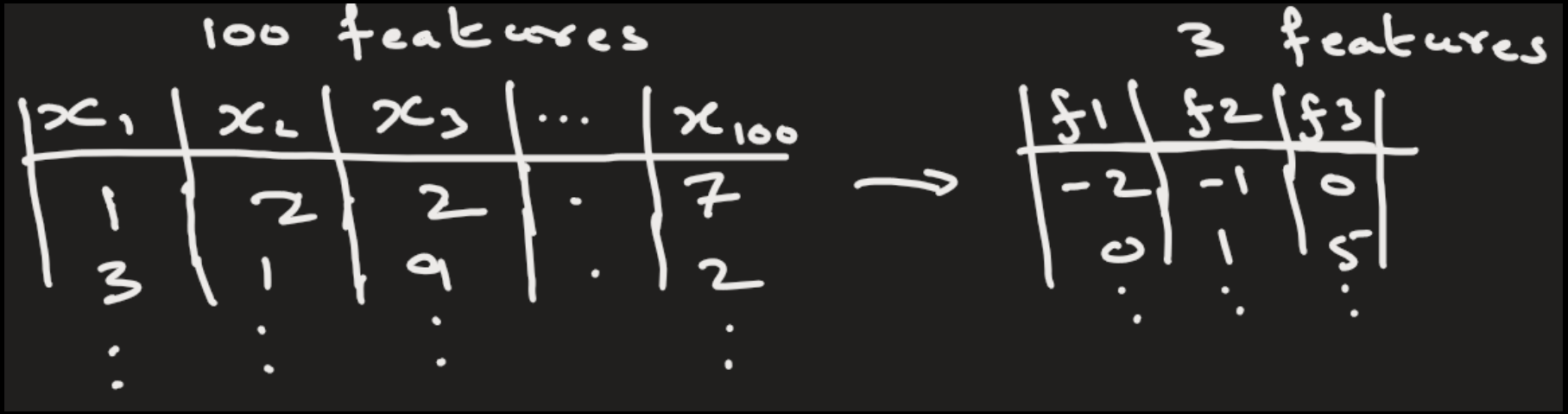
Reduces number of features while preserving important information.

It helps in

- Data compression – Less features means less data
- Visualization – easy to see 2D or 3D
- Noise reduction – remove unnecessary columns

Common Algorithms:

- PCA
- t-SNE
- UMAP
- Autoencoders



Example:

Visualizing 100-dimensional data in 2D

Reducing features before ML modeling

Types

Association Rule Learning

It finds relationships between variables.

The goal is to discover "if-then" patterns.

Common Algorithms:

- Apriori
- FP-Growth
- Eclat

Example:

Market Basket Analysis

If customer buys bread → also buys butter

If students buys notebooks → also buys pen

If month is May, June, July, August (Summer)-> More consumption of cold drinks

Types

Anomaly Detection

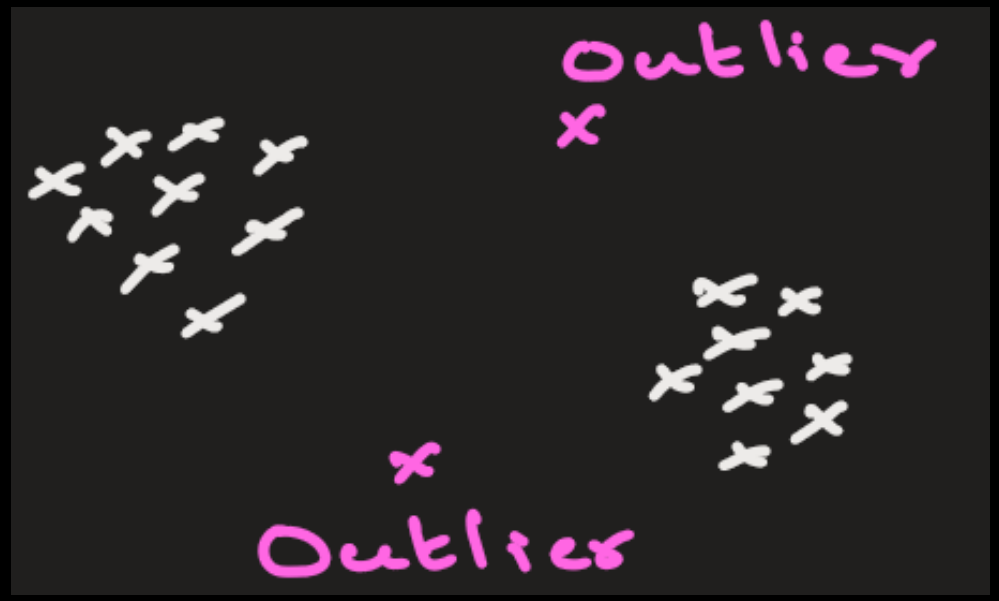
It identifies rare or unusual data points, i.e. outliers.

Common Algorithms:

- Isolation Forest
- One-Class SVM
- Local Outlier Factor

Example:

- Fraud detection
- Network intrusion detection
- Fault detection



Types

Generative Models:
Learns data distribution and generates new samples.

Common Models:
- GANs
- Variational Autoencoders (VAE)

Example:
- Generate new images
- Synthetic data creation: 100 images -> 200 images

Summary

Type	Main Purpose	Example Use Case
Clustering	Group similar data	Customer segmentation
Dimensionality Reduction	Reduce features	Visualization
Association Rules	Find relationships	Market basket analysis
Anomaly Detection	Detect rare events	Fraud detection
Generative Models	Create new data	Image generation

Heading Goes Here

Fhdsklf
Fjdsklf
Fjskldf